

seL4/CAmkES Reference

Page Content

- [CAmkES terminology](#)
 - [Building blocks](#)
 - [Full system description](#)
 - [Component description](#)
 - [Connector description](#)

CAmkES terminology

Building blocks

Full system description

An **assembly** marks a complete system description, consisting of

- a **composition** and
- an optional **configuration**

A composition is a container for the **component** and **connector** instantiations that form the actual system. A configuration is a container for describing **settings**, which hold the assignment of **attributes**.

```
assembly {  
  composition {  
    component foo f;  
    component bar b;  
  
    connection RPC c1(from f.t1, to b.t2);  
    connection Notification c2(from f.s1, to b.s2);  
    connection SharedData c3(from f.b1, to b.b2);  
  }  
  configuration {  
    f.kitty = {"name": "meows", "paws": 4};  
  }  
}
```

Component description

A component can contain specific parameters, which allow for different functionality

```
component foo {  
  control;  
  uses thing t1;  
  emits sig s1;  
  dataport buffer b1;  
  attribute cat kitty;  
}  
  
component bar {  
  provides thing t2;  
  consumes sig s2;  
  dataport buffer b2;  
}
```

Keywords describe the type and behavior of a component:

- **control**: the component is an active component that possesses an active thread of execution
- **hardware**: the component represents a hardware device

```
struct cat {
    int paws;
    string name;
}
```

Attributes provide extra data of an arbitrary type, and are used to specialise or differentiate a component at runtime. According to the CAMkES manual, the description of a component has to describe the name of the attribute and its type. The types can be constructed as a **struct** (e.g. **cat**) of any of the basic CAMkES types:

- (unsigned) int
- (unsigned) char
- string

Dataports (e.g. **b1**) mark a component's port (e.g. **buffer**) interfaces and are represented as shared memory regions.

keywords

- maybe
- has
- dataport
- event
 - **consumes** (receive/handle)
 - **emits** (produce)
- interface
 - **provides**: Procedure interfaces implemented by a component
 - **uses**: Procedure interfaces that are invoked by a component
- **connection**: instantiation of the **connector**, connects two instances

Attributes provide extra data of an arbitrary type. According to the CAMkES manual, the description of a component/connector must describe the name of the attribute and its type.

- connector: type
- attribute (name + type)

- dataport: port interfaces
- shared memory

- direction: the flow of parameter of a procedure method
- in (caller to the callee)
- out (callee to the caller)
- inout (bidirectional)
- refin (identical to 'in' except for the C backend where it is optimised to pass-by-reference)

- event: the asynchronous signal interface of a component
- defined by an identifier (numerical value)

- instance: instantiation of a component type

- interface: abstract exposed interaction point of a component
- maybe: keyword, declaring interface as optional; functions/dataport pointers associated are NULL at runtime
- interface type:
- port: represents shared memory semantics

- procedure: interface with function call semantics
- method: item of a procedure (maps to a function)
- direction
- type: return/parameter type (e.g. string)
- parameter: a piece of data referenced by a procedure method (argument to a function)
- setting: instantiation of attribute(s)

Connector description